Chemical Reactions and Equations — MCQ Worksheet

Science • Chapter 1 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Which of the following is NOT a sign that a chemical reaction has taken place?

(A) Change in colour
(B) Change in temperature
(C) Change in shape due to cutting
(D) Evolution of a gas

Q2. When magnesium ribbon burns in air, the white powder formed is:

(A) Magnesium hydroxide
(B) Magnesium oxide
(C) Magnesium carbonate
(D) Magnesium sulphate

Q3. A balanced chemical equation obeys the:

(A) Law of definite proportions
(B) Law of multiple proportions
(C) Law of conservation of mass
(D) Law of constant composition

Q4. In the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$, the gas evolved is identified by:

(A) Limewater turning milky
(B) A burning splinter making a 'pop' sound
(C) Brown fumes formation
(D) A pungent smell

Q5. Which of the following is a combination reaction?

(A) $2H_2O_2 \rightarrow 2H_2O + O_2$
(B) $CaO + H_2O \rightarrow Ca(OH)_2$
(C) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
(D) $NaOH + HCl \rightarrow NaCl + H_2O$

Q6. Decomposition of ferrous sulphate on heating produces:

(A) $FeO + SO_2$
(B) $Fe_2O_3 + SO_2 + SO_3$
(C) $Fe + S + O_2$
(D) $Fe(OH)_3 + SO_3$

Q7. When silver chloride is exposed to sunlight, it turns grey because:

(A) It absorbs moisture
(B) It is photolytically decomposed to silver and chlorine
(C) It reacts with oxygen in air
(D) It dissolves in air

Q8. An iron nail dipped in copper sulphate solution turns brown because:

(A) Iron is more reactive than copper, so copper is displaced
(B) Copper is more reactive than iron
(C) The solution gets diluted
(D) Iron sulphate is brown

Q9. Which of the following is a double displacement reaction?

(A) $Fe + S \rightarrow FeS$
(B) $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$
(C) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
(D) $2H_2O \rightarrow 2H_2 + O_2$

Q10. Oxidation is best defined as:

(A) Loss of oxygen
(B) Loss of hydrogen or loss of electrons
(C) Gain of hydrogen
(D) Gain of electrons

Q11. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance reduced is:

(A) CuO
(B) H_2
(C) Cu
(D) H_2O

Q12. Respiration is an example of:

(A) Endothermic reaction
(B) Exothermic reaction
(C) Photolytic reaction
(D) Electrolytic reaction

Q13. Photosynthesis is a type of:

(A) Exothermic reaction
(B) Endothermic reaction
(C) Combination reaction only
(D) Decomposition reaction only

Q14. The reddish-brown coating formed on iron when exposed to moist air is:

(A) Fe_2O_3
(B) FeO

(C) Hydrated Fe_2O_3 (rust) (D) $Fe(OH)_2$

Q15. Galvanisation is the process of coating iron with:

(A) Tin
(B) Copper
(C) Zinc
(D) Aluminium

Q16. Which gas is filled in chip packets to prevent rancidity?

(A) Oxygen
(B) Carbon dioxide
(C) Nitrogen
(D) Hydrogen

Q17. The chemical formula of slaked lime is:

(A) CaO
(B) $Ca(OH)_2$
(C) $CaCO_3$
(D) $CaCl_2$

Q18. When CO_2 is passed through limewater, it turns milky due to formation of:

(A) Calcium oxide
(B) Calcium hydroxide
(C) Calcium carbonate
(D) Calcium bicarbonate

Q19. Which of the following is an electrolytic decomposition reaction?

(A) $2AgCl \rightarrow 2Ag + Cl_2$
(B) $2H_2O \rightarrow 2H_2 + O_2$ (using electricity)
(C) $CaCO_3 \rightarrow CaO + CO_2$
(D) $2HgO \rightarrow 2Hg + O_2$

Q20. The substance which gets oxidised in a redox reaction is called the:

(A) Oxidising agent
(B) Reducing agent
(C) Catalyst
(D) Inhibitor

Q21. Balanced form of the equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ is:

(A) $Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$
(B) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
(C) $2Fe + 3H_2O \rightarrow Fe_3O_4 + 3H_2$
(D) $3Fe + 2H_2O \rightarrow Fe_3O_4 + H_2$

Q22. The state symbol used for an aqueous solution in a chemical equation is:

(A) (s)
(B) (l)
(C) (g)
(D) (aq)

Q23. Antioxidants are added to packaged food to:

(A) Improve flavour
(B) Prevent rancidity
(C) Add colour
(D) Increase weight

Q24. Which of the following changes is NOT a chemical change?

(A) Burning of paper
(B) Rusting of iron
(C) Melting of ice
(D) Curdling of milk

Q25. When dilute HCl is added to sodium carbonate, the gas evolved is:

(A) Hydrogen
(B) Oxygen
(C) Carbon dioxide
(D) Sulphur dioxide

— End of Worksheet —